HARSHDEEP PLAHA

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EDUCATION

University of Missouri - Kansas City **December 2024**
Bachelor's, Computer Science *GPA: 3.83*

PROFESSIONAL EXPERIENCE

Bliink AI Payments **Iselin, NJ, USA**
Data Scientist *January 2025 - Present*

- Architected cloud-native data platform on Kubernetes (AKS) with Kafka CDC connectors, orchestrating 10+ Airflow DAGs for ETL through Bronze-Silver-Gold medallion architecture using Apache Iceberg, Nessie, and ADLS2.
- Engineered production feature store serving 150+ fraud/AML features with sub-10ms latency using Feast, Apache Spark, and two-tier architecture (Iceberg offline + Redis online) with automated materialization pipelines.
- Automated ML training and retraining pipelines reducing model refresh time by 70% using Spark clusters and MLflow for experiment tracking, versioning, and model registry with scheduled workflows.
- Deployed scalable model serving infrastructure achieving sub-100ms P99 latency for real-time fraud inference on Kubernetes, handling thousands of concurrent scoring requests in production.
- Built investigation agents using fine-tuned Phi-3 with LangChain for tool orchestration; implemented LangSmith for end-to-end traceability of agentic workflows in production AML/fraud systems.
- Reduced analyst triage time by 65% by engineering GenAI case-resolution agent that retrieves context from Neo4j, PostgreSQL, and Feast to generate tier-1 risk summaries and arbitration recommendations.

Q2 **Austin, TX, USA**
Software Engineer Intern *May 2024 - August 2024*

- Upgraded web services [Adapters] within distributed systems using Kessel, Q2's proprietary Python-based request handling framework for its microservices moving over 45 financial institutions to the latest technology, to Kessel 2.0 following Agile SDLC working with cross functional teams.
- Enhanced system functionality, observability, and monitoring capabilities following design patterns by implementing Kessel 2.0 methods and new event-driven logging features using advanced internal logging mechanisms and REST APIs to effectively manage json data.
- Gained familiarity with container orchestration and the CI/CD deployment process, identified necessary changes in deployment configs Terraform (HCL files) for the switch to Kessel 2.0, then leveraged Jenkins and Nomad for automated DevOps workflows. Used bitbucket for version control
- Ran unit tests following test driven development for all adapters after implementing upgrades locally, followed by end-to-end testing in remote environments to ensure seamless functionality and integration of services across distributed systems.

University of Missouri Kansas City **Kansas City, MO, USA**
Research Assistant *October 2023 - July 2024*

- Enhanced data mining capabilities by implementing a data harvesting solution that utilized Python and Airflow to collect over 4 TB of meteorological data, achieving approximately 500 GB/day from NOAA, MERRA-2, and CEDS APIs while supporting distributed storage.
- Built a ETL pipeline for the meteorological data on remote HPC servers using apache spark, interpolating to 1 mile grids, generating 60+ engineered features, and surfacing top predictors (e.g., NOx, specific humidity) for ozone-forecast models
- Partnered with the University of Miami to build Unity-based VR simulations for child-behavior research; implemented immersive scene logic and developed an NLP sentiment-analysis pipeline boosting participant engagement by 25 %.

HID **Kansas City, MO, USA**
Data Collection Assistant *September 2024 - December 2024*

- Captured and curated 12K+ synchronized RGB frames, depth maps, and IR images with HID's U.R.U. 3D face cameras under Dr. Reza Derakhshani, expanding the multimodal training corpus for HID Global's nextgen face recognition engine by 35%.
- Coordinated and executed 300+ participant sessions, administering digital consent, enforcing IRB and dataprivacy safeguards, and encrypting biometric files maintained 100% regulatory compliance with zero privacy incidents.
- Engineered a Pythonbased ETL pipeline that ingested, validated, and cataloged the RGB + depth + IR image streams into secure object storage with metadata, automating checksum validation and quality scoring to cut preprocessing time and provide researchers with analyticsready biometric datasets.

PROJECTS

Horizon

- Developed a full-stack GenAI financial SaaS platform that connects and aggregates data from multiple bank accounts via Plaid, providing a unified financial overview, expense tracking, and savings recommendations powered by a locally hosted Meta LLaMA 3 7B Instruct model through Ollama.
- Fine-tuned LLaMA 3 7B Instruct on anonymized user financial data to specialize it for banking transactions, improving context comprehension and classification accuracy by ~41 %, and enabling 92 % accurate transaction categorization across 15+ spending categories.
- Delivered an interactive React/Chart.js dashboard with sub-second inference latency, allowing users to query spending trends in natural language, visualize financial progress

VR Simulation for Child Behavior Analysis *October 2023 - January 2024*
Research Assistant

- Built and designed dynamic scenes for various scenarios, implementing game logic in C# using .NET and Visual Studio Code to create interactive and responsive environments, aligning with the need for geometric and semantic representation of real-world spaces in AR/MR applications.
- Developed a real-time speech-to-sentiment analysis pipeline using hugging face API and Meta voice SDK that enhanced character interactivity by enabling natural language processing and emotional response analysis, contributing to improved user engagement and immersive experiences.
- Integrated multi-view geometric modeling techniques to ensure accurate representation of different camera viewpoints, improving the realism and depth of the VR simulation.

SKILLS

- **Languages:** C/C++, Python, R, C#, Java, TypeScript, JavaScript, SQL, HTML/CSS, Shell Scripting, GQL
- **Databases:** MySQL, Qdrant, Neo4J, Postgres
- **Frameworks and Libraries:** NumPy, Pandas, Tensorflow, Keras, Scikit Learn, Pytorch, React.js, OpenCV, Node.js, Hugging Face, Apache Kafka, Fast api, pyspark, React Native, LangChain, FastMCP
- **Tools:** Kubernetes, Git, Jenkins, Docker, Linux/Unix, Apache Spark, REST APIs, Atlassian, Teradata, AWS, Apache Airflow, Microsoft Azure, GCP, Apache Spark, Apache Iceberg, mlflow, kubeflow, LangSmith